Evaluating a SQL explainer with a golden set and an LLM judge

Project 1 · CS496 · 2026-10-21 · system prompt v3 on `gpt-5-nano` · judge v1 on `gpt-5-mini` (OpenAI) · details in README, CHANGELOG and postmortem

> **Result.** System v3 is rated good on **77.2%** of 101 test queries (mean of 3 runs: 78, 76, 80), up from about 69% for the baseline prompt on dev. The judge that produces this score agrees with our golden labels on only **64% of test items (κ 0.33)**. It catches 93% of bad answers but fails about a third of good ones, so treat 77% as a judge-graded estimate that is likely on the low side. Running the system costs **$0.19 per 1,000 requests**.

Task and golden set

The system explains a SQLite query in plain words for a reader who doesn't know SQL. An explanation is good only if it is correct, complete, invents nothing and is clear. Word overlap can't grade this: on c-q117, two good answers share 42% of their words, while a wrong one ("skips the top two prices") shares 97% with a good one. We built 161 queries (60 dev / 101 test), each with one blind candidate: 73 baseline, 32 from a second model, and 56 with planted errors of six types. Two Claude labellers agreed on **157/161 (97.5%, κ 0.95)**. After adjudicating 4 disagreements and making 17 rule-based overrides, the set is **82 good / 79 bad**.

Judge reliability

Judge v1 vs golden labels	Dev (60)	Test (101)
Agreement / Cohen's κ	54/60 (90%) / 0.79	**65/101 (64%) / 0.33**
Bad answers caught / "bad" calls that are right	34/36 / 34/38	40/43 / 40/73
Agreement on hard queries	86%	39%

- **Dev overstates reliability.** We fixed dev labels wherever the judge exposed a rule violation, but not test labels. Test disagreements repeat the patterns we fixed on dev (e.g. c-q035, c-q156). The judge is also too strict: about 23 of its 33 wrong "bad" calls demand details the guide says may be left out.
- **It catches planted errors** (34/35 on test) but misses clause-by-clause SQL jargon (c-q120, c-q140).
- **No position bias:** same verdict in both orders on 87/101 pairs; first slot won 30%, second 31%.
- **No verbosity bias:** padding an answer to about 2.8× its length, with the content unchanged, rescued 2/40 bad answers, and the judge preferred the original 35 to 25.

System prompt history

gpt-5 models take no temperature, so two identical runs differ by about 3 points (19 items changed grade on dev), and we compared versions across reruns.

Version	Change	Dev	Test
v1	Baseline: "explain in plain words"	43, 40 / 60	—
v2	Be exact about which rows are included (joins, filters)	43 / 60	—
v3	v2 + describe row order only from `ORDER BY`, with its direction	**51, 47 / 59**	**78, 76, 80 / 101**

v2's changes were within noise. v3 beat v1 in every run. Hard queries remain the weak spot (62% on test).

Cost

Per item: system $0.00019 (3.7 s median), judge $0.0021 (10 s median). Per 1,000 requests: **$0.19**, or **$0.39** if the judge grades a 10% sample. At an assumed 500 requests/day that's about $5.90/month, and about $588/month at 100×. The whole evaluation cost about $4 on OpenAI.

Main lessons

- Measure the judge on a split whose labels it never influenced. Our dev agreement figure was inflated.
- With models that ignore temperature, one run isn't a measurement. Rerun before you believe a change.